4. c) molecular solid

lattice points are occupied by CO_2 molecules

5. a) Both assertion and reason are true and reason is the correct explanation of assertion.

6. c) 8 and 4

CaF_2 has cubical close packed arrangement

Ca^{2+} ions are in face centered cubic arrangement, each Ca^{2+} ions is surrounded by 8 F^- ions and each F^- ion is surrounded by 4 Ca^{2+} ions.

Therefore coordination number of Ca^{2+} is 8 and of F^- is 4

7. b) 6.023 X 1022

in bcc unit cell,

2 atoms $\equiv$ 1 unit cell

Number of atoms in 8g of element is ,

$$\text{Number of moles} = \frac{8g}{40\ \text{g mol}^{-1}} = 0.2\ \text{mol}$$

1 mole contains 6.023×10^{23} atoms

0.2 mole contains $0.2\times6.023\times10^{23}$ atoms

$$\left(\frac{1\text{unit cell}}{2\ \text{atoms}}\right)\times0.2\times6.023\times10^{23}$$

6.023×10^{22} unit cells

8. a) 8

in diamond carbon forming fcc. Carbon occupies corners and face centres and also occupying half of the tetrahedral voids.

$$\left(\frac{N_c}{8}\right)+\left(\frac{N_f}{2}\right)+4\text{ C atoms inTd voids}$$

$$\left(\frac{8}{8}\right)+\left(\frac{6}{2}\right)+4=8$$

9. d) M_3N_2

if the total number of M atoms is n, then the number of tetrahedral voids =2n

given that $\left(\frac{1}{3}\right)^{rd}$ of tetrahedral voids are occupied i.e., $\left(\frac{1}{3}\right)\times2n$ are occupied by N atoms

$$\therefore \text{M:N} \Rightarrow n:\left(\frac{2}{3}\right)n$$

$$1:\left(\frac{2}{3}\right)$$

$$3:2 \Rightarrow M_3N_2$$

10. b) 15.05%

let

the number of Fe^{2+} ions in the crystal be x

the number of Fe^{3+} ions in the crystal be y

total number of Fe^{2+} and Fe^{3+} ions is $x+y$

given that $x+y=0.93$

the total charge =0

$$x(2+)+(0.93-x)(+3)-2=0$$

$$2x+2.97-3x-2=0$$

$$x=0.79$$

Percentage of Fe^{3+}

$$=\left(\frac{(0.93-0.79)}{(0.93)}\right)100=15.05\%$$

11. c) 6

$$\frac{r_{C^+}}{r_{A^-}}=\frac{0.98\times10^{-10}}{1.81\times10^{-10}}=0.54$$

it is in the range of 0.414 0.732 ,hence the coordination number of each ion is 6

12. d) $\left(\frac{\sqrt{3}}{2}\right)\times400\text{pm}$

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